

IS BROADER BETTER? A CONTROLLED STUDY OF MULTILINGUAL COVERAGE AND PRETRAINING OBJECTIVE IN FROZEN SSL ENCODERS FOR SPEECH DEEPFAKE DETECTION

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ABSTRACT

Frozen self-supervised (SSL) speech encoders are strong, low-cost front ends for audio deepfake detection, and recent comparisons agree that large, multilingual, discriminative encoders generalize best out of domain. These comparisons fail to control for encoder capacity, pretraining objective, and multilingual coverage together, identifying *which* encoder wins without isolating *why*. We present a controlled decomposition with a fixed pipeline and trainable capacity. We vary multilingual coverage on four wav2vec2-family encoders, matched to ~ 315 M parameters. We isolate the pretraining objective on two encoders matched on identical data. Coverage does not help monotonically, as out-of-domain error drops sharply at the ~ 100 -language scale (XLS-R) but does not improve further at the 1406-language extreme (MMS). We find that a mid-coverage encoder is strongest on farther out-of-domain sets, matching or surpassing a 577M-parameter model at 315M. Its lead on these far sets, statistically significant under paired bootstrap, and on the official ASVspoof 5 cost metric holds under two backends. Separately, masked-prediction pretraining generalizes better than contrastive on identical data (In-the-Wild EER 26.5% vs. 46.8%). Within this fixed frozen-encoder recipe, we find that broader and larger models are not reliably better.

Index Terms— audio deepfake detection, anti-spoofing, self-supervised learning, frozen encoders, multilingual pretraining

1. INTRODUCTION

Self-supervised learning (SSL) speech encoders are now the dominant front end for audio deepfake detection [1], and recent work shows that even *frozen* encoders paired with a lightweight backend generalize well to out-of-domain (OOD) attacks at a fraction of the training cost [2, 3]. This makes the choice of frozen encoder a fundamental design decision: the representation is fixed, so whatever it fails to capture cannot be recovered downstream or in deployment. The property of an encoder’s pretraining that drives OOD generalization remains unresolved.

Recent comparisons report that large, multilingual, discriminative encoders such as XLS-R generalize best [3, 4], while concurrent studies question whether scale alone explains this [5, 6]. But these comparisons vary encoder capacity, pretraining objective, and multilingual coverage *simultaneously* (Spoof-SUPERB spans ~ 4 M–317M parameters [3], and MMS is evaluated at 1B [4, 5]), so they establish *which* encoder wins without isolating *why*. The advantage could reflect scale, objective, coverage, or their interaction. The conundrum is unresolved because prior comparisons rarely hold the other factors fixed.

To our knowledge, we provide the first controlled decomposition of coverage and objective at matched capacity. Fixing trainable-backend capacity, pretraining objective, and pipeline, we sweep multilingual coverage across four wav2vec2-family encoders matched to ~ 315 M parameters (monolingual wav2vec2-LV60, 53-language XLSR-53, 128-language XLS-R, and 1406-language MMS-300M), with coverage, and its correlated data volume, as the varying factor. Our central finding is that coverage does not help monotonically as out-of-domain error drops sharply once coverage reaches the ~ 100 -language scale (XLS-R), and does not improve at the 1406-language extreme (MMS). The 128-language encoder is the strongest choice on the farther out-of-domain sets while both endpoints are worse. XLS-R is also best on the official ASVspoof 5 cost metric under two independent backends (0.49–0.56, vs. 0.60–0.65 for MMS and ≥ 0.87 for XEUS and the English-only encoders). At 315M parameters, it also matches or outperforms the 577M-parameter XEUS out of domain as one data point against parameter scale, rather than pretraining coverage, as the driver. We further isolate the pretraining *objective*, a factor prior comparisons did not isolate, by contrasting two encoders trained on *identical* data (wav2vec2 vs. HuBERT, both on Libri-Light-60k, both ~ 315 M): masked prediction generalizes markedly better than the contrastive objective at fixed (monolingual) coverage. Together, these controlled comparisons indicate that OOD generalization depends more on an encoder’s pretraining — its multilingual coverage and objective — than on its parameter count, and that broader coverage is not always better.

Our contributions are: (i) to our knowledge, the first matched-capacity sweep of multilingual coverage across multiple coverage levels (1 to 1406 languages) for frozen SSL anti-spoofing, revealing a non-monotonic coverage effect that peaks near 100 languages; (ii) a matched-data isolation of pretraining objective, finding masked prediction generalizes better out of domain than contrastive learning; and (iii) multi-seed evaluation with paired-bootstrap significance testing across five benchmarks and two backends, including the official ASVspoof 5 cost metric, in contrast to prior single-run comparisons.

2. RELATED WORK

SSL front ends underpin most anti-spoofing systems. Early work established their value with fine-tuned encoders [7, 1], and recent systems show frozen encoders with lightweight backends generalize well at far lower cost [2]. A parallel line compares encoders to identify which generalize best. Spoof-SUPERB [3] benchmarks 20 frozen SSL models under a unified protocol and finds large multilingual discriminative encoders strongest, attributing this to a bundle of scale, multilingual pretraining, and speaker-aware objectives; con-

current preprints question whether scale alone explains robustness [5, 6]. Chetia Phukan et al. [4] likewise find multilingual encoders outperform monolingual ones, but compare 1B-parameter multilingual models (XLS-R, MMS) against sub-350M monolingual ones, conflating coverage with a $\sim 3\text{--}10\times$ capacity gap, and treat multilinguality as binary, leaving open whether *more* coverage continues to help. Notably, when trained on ASVspoo 2019 and tested on In-the-Wild, as in our setting, MMS outperforms XLS-R in their cross-corpus evaluation, opposite to our matched-capacity result; their use of 1B checkpoints, final-layer features, and a different downstream pipeline may account for the discrepancy. These comparisons, in general, vary encoder capacity, objective, and coverage together ([4] and [5] evaluate MMS at 1B, while SpooF-SUPERB spans $\sim 4\text{M--}317\text{M}$ parameters), establishing “winning” encoders without isolating why. RAPTOR [6], a controlled study at $\sim 100\text{M}$ parameters, similarly attributes robustness to multilingual pretraining rather than scale, convergent with our findings at a different capacity band. We provide the matched-capacity and matched-data controls these comparisons lack. By sweeping coverage at fixed capacity, we directly test whether a broader model is better.

3. METHOD

3.1. Frozen-encoder pipeline

We have a fixed detection pipeline across all conditions, isolating the frozen encoder as the only variable. Each input waveform is passed through a frozen SSL encoder. We take the mean of the encoder’s last four transformer layers as the frame-level representation, following evidence that SSL layers encode different information [8] and that spoofing cues vary by layer [9].¹ In parallel, a constant-Q cepstral (CQCC) branch provides a spectral stream. Both streams are projected to $D=128$ and fused by the cross-attention recipe of [2], followed by a backend (Section 3.4) and a linear bona-fide/spoof classifier. Only the projections, fusion, backend, and classifier are trained. The encoder is frozen throughout. Fusion uses 4-head cross-attention (dropout 0.1) while the MHFA backend uses $K=64$ compression, $H=8$ heads, and 128-dim embeddings (dropout 0.2). All systems are trained on ASVspoo 2019 LA with no augmentation, over three seeds {42, 1234, 2025}, selecting the checkpoint by best LA19-dev EER. We use AdamW with two learning rates (3×10^{-4} for the backend, 1×10^{-4} for the fusion front end), weight decay 10^{-4} , gradient clipping 1.0, batch size 32, StepLR (halved every 10 epochs), and a weighted-cross-entropy loss (bona-fide upweighted) with label smoothing 0.1. MHFA cells train for 30 epochs; AASIST cells for 40, warm-started from the matched MHFA head. The CQCC branch uses 60-dim features (20 coefficients with Δ , $\Delta\Delta$).

3.2. Scale normalization as a control

Raw SSL features differ by roughly two orders of magnitude across encoders (RMS ≈ 770 for XLSR-53 vs. ≈ 5 for MMS). Fed directly to a projection with fixed initialization and learning rate, this makes each encoder’s first layer train at a $\sim 100\times$ different *effective* learning rate, confounding encoder identity with input scale, precisely the factor our study must control. We therefore apply a scale-matching normalization to the frozen features before projection: a non-affine LayerNorm followed by a fixed scalar s , identical for every encoder. We set $s \approx 700$ to match the mean pre-normalization SSL magnitude,

preserving the fusion’s original SSL-to-spectral balance. s is a scale-matching constant, *not* a tuned hyperparameter. This removes cross-encoder scale confounding. As a validation that the normalization behaves as intended, it also resolves a bimodal in-domain instability we observe for XLSR-53 (seed std $16.68 \rightarrow 1.63$). Crucially, because the non-affine LayerNorm maps every encoder to unit variance regardless of s , the *cross-encoder* scale matching, our actual fairness mechanism, is exact for any $s > 0$. s affects only the shared SSL-to-spectral branch balance, which we fix to the anchor recipe’s. The encoder comparison is therefore robust to the choice of s by construction, not by tuning.

3.3. Two controlled axes

We isolate two pretraining factors by varying each while holding the other fixed.

Coverage. Four wav2vec2-family encoders (monolingual wav2vec2-LV60 [10], XLSR-53 [11] (53 languages), XLS-R [12] (128), and MMS-300M [13] (1406)) share the same contrastive objective, architecture (24 layers, 1024-dim), and $\sim 315\text{M}$ parameters, and differ primarily in multilingual coverage. One caveat is irreducible, however, as coverage co-varies with pretraining data *volume* ($\approx 50\text{K}$ to $\approx 500\text{K}$ hours) across public checkpoints, which cannot be separated without controlled pretraining. We therefore attribute effects to coverage and its correlated data scale, not to language count alone.

Objective. wav2vec2-LV60 and HuBERT-LARGE [14] are both pretrained on *identical* Libri-Light-60k data at $\sim 315\text{M}$ parameters, differing in objective (contrastive vs. masked prediction). The pre-training data is matched exactly, making this a clean contrast with respect to data.

3.4. Capacity, reference, backends, metrics

All coverage and objective encoders are matched at 315.4M parameters.² We include the 577M massively-multilingual XEUS [15] as a labeled *reference*, a larger, broader encoder outside the matched set, to ask whether matched-capacity coverage can rival it. We evaluate two backends, MHFA [16] and AASIST [17], on five benchmarks, each of increasing distance from training: LA19 (in-domain) [18], LA21, DF21 [19], In-the-Wild (ITW) [20], and ASVspoo 5 (ASV5) [21]. We report pooled EER as the mean over three seeds (seed std where relevant) and the official ASVspoo 5 countermeasure min-DCF ($P_{\text{spoo}}=0.05$) on the large-scale set. We build on the cross-attention SSL/spectral fusion recipe of [2].

4. RESULTS

Table 1 reports pooled EER and ASVspoo 5 min-DCF for both controlled axes under the MHFA backend, with the 577M XEUS reference.

4.1. Coverage helps up to a point

With capacity and objective held fixed, OOD error is *not* monotonic in multilingual coverage. The two higher-coverage encoders (128-language XLS-R and 1406-language MMS-300M) dramatically outperform both the monolingual endpoint (wav2vec2-LV60) and the 53-language XLSR-53 on nearly every OOD benchmark by EER

¹For XEUS (19 blocks) we average blocks 16–19, matching the last-four convention.

²Exact counts: wav2vec2-LV60, XLSR-53, XLS-R, and MMS-300M are 315.4M each (24 transformer layers, 1024-dim); HuBERT-LARGE 315.4M; WavLM-LARGE 315.5M. XEUS is 577M.

Table 1. Pooled EER (% , mean over 3 seeds) and official ASVspoof 5 min-DCF, MHFA backend. Coverage axis (matched $\sim 315M$, wav2vec2 objective) and objective axis (matched Libri-Light-60k data). XEUS is a 577M reference outside the matched set. **Bold:** best per column among matched encoders. † WavLM adds a denoising objective *and* more/broader data (94k h), so it is not a clean same-data contrast.

Encoder	LA19	LA21	DF21	ITW	ASV5	mDCF
<i>Coverage (matched $\sim 315M$, contrastive)</i>						
wav2vec2-LV60 (1)	3.52	27.15	18.13	46.77	41.88	0.94
XLSR-53 (53)	3.67	30.32	22.90	38.97	25.40	0.65
XLS-R (128)	1.41	8.50	8.34	18.30	21.18	0.56
MMS-300M (1406)	1.43	11.19	8.48	22.71	24.36	0.65
<i>Objective (matched LL-60k data, $\sim 315M$)</i>						
wav2vec2-LV60 (contr.)	3.52	27.15	18.13	46.77	41.88	0.94
HuBERT (masked)	1.09	13.88	10.96	26.54	38.40	1.00
WavLM (masked+denoise) †	0.92	11.75	12.69	46.03	43.96	1.00
<i>Reference (577M, outside matched set)</i>						
XEUS	0.66	9.04	13.83	30.30	33.96	0.97

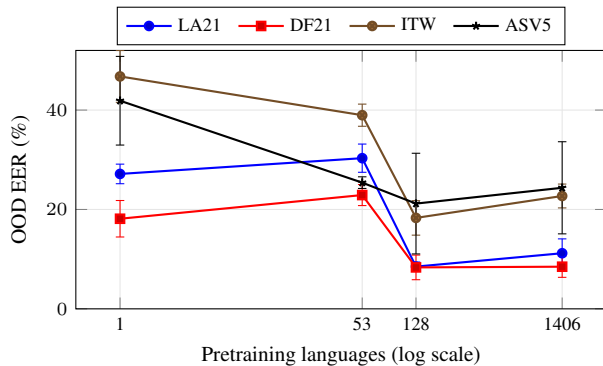


Fig. 1. Out-of-domain EER vs. multilingual coverage (MHFA, mean over 3 seeds, error bars ± 1 std), for the four matched-capacity ($\sim 315M$) wav2vec2-family encoders. Every benchmark bottoms out at the 128-language XLS-R, then rises again at MMS: coverage helps up to a point, not monotonically. Seed variance is large on ASVspoof 5, where we rely on the more stable min-DCF (text).

(the cost-metric picture is qualified below). Among the two, XLS-R has the lower EER on the two farther sets (ITW 18.30 vs. 22.71, ASV5 21.18 vs. 24.36) and on LA21 (8.50 vs. 11.19), and is level with MMS on DF21 (8.34 vs. 8.48). The trend is also non-monotonic at the low end: the 53-language XLSR-53 is *worse* than the monolingual encoder on LA21 (30.32 vs. 27.15) and DF21 (22.90 vs. 18.13). OOD error therefore does not fall steadily with language count. It drops sharply once coverage reaches the ~ 100 -language scale (XLS-R) but shows no further gain at the 1406-language extreme, where it degrades by the cost metric below. Figure 1 shows this shape directly, as every out-of-domain benchmark bottoms out at the 128-language XLS-R.

Because several encoders show high seed variance in EER on the large-scale ASVspoof 5 set (XLS-R 21.18 ± 10.13 , driven by one outlier seed of three), the EER ranking there is seed-fragile, so we treat the more stable official ASVspoof 5 min-DCF as the primary metric on that set. On min-DCF, XLS-R attains the best cost under *both* backends (0.56 MHFA, 0.49 AASIST), ahead of MMS (0.65,

0.60) and far ahead of the English-only encoders, whose min-DCF is near-degenerate on this set (0.94–1.00). XLS-R’s cost-metric lead under two architecturally distinct backends, together with the consistent gap between high and low coverage encoders, is the most stable form of the ranking. On ASV5 min-DCF, XLSR-53 and MMS are level (both 0.65), so the tier gap separates XLS-R from the rest rather than splitting cleanly into two tiers.

The pattern also holds against a substantially larger model. Out of domain, matched-capacity XLS-R (315M) matches or surpasses the 577M XEUS on the OOD benchmarks (most sharply on ITW 18.30 vs. 30.30, ASVspoof 5 21.18 vs. 33.96, and cost metric 0.56 vs. 0.97), despite roughly half the encoder size. In domain (LA19), XEUS is strongest (0.66 vs. 1.41), so the advantage is specifically one of out-of-domain generalization rather than overall capability.

Significance. We test the key comparisons with a paired bootstrap over evaluation utterances (class-stratified, 1000 replicates, MHFA; the resampling unit is the utterance with the three trained models held fixed, so intervals reflect evaluation-set sampling, not training-seed variance, which we report separately). XLS-R’s advantage over MMS is significant on LA21, ITW, and ASV5 (95% CIs excluding zero, all three seeds agreeing) but *not* on DF21 ($\Delta = -0.13$, CI $[-0.38, +0.13]$, MMS ahead on two of three seeds), so the two higher-coverage encoders are statistically level on the near sets and separated on the far ones (ITW -4.42 $[-4.81, -4.03]$; ASV5 -3.19 $[-3.30, -3.07]$). XLS-R’s advantage over XEUS is significant on all four OOD sets and largest on ITW (-12.0 $[-12.5, -11.6]$) and ASV5 (-12.8 $[-12.9, -12.7]$). On ASV5 the bootstrap confirms the *direction* of these gaps (all three seeds agree) but not their magnitude: XLS-R’s per-seed EER spans 15.0–32.9, so the utterance-level interval is much narrower than the training-seed spread, and we rely on min-DCF for the ASV5 ranking accordingly.

4.2. Pretraining objective matters independently

The objective contrast isolates a second factor. On *identical* Libri-Light-60k data at matched capacity, the masked-prediction encoder (HuBERT) generalizes markedly better than the contrastive one (wav2vec2-LV60): LA21 13.88 vs. 27.15, DF21 10.96 vs. 18.13, ITW 26.54 vs. 46.77, and in-domain 1.09 vs. 3.52. Because pretraining data is matched exactly, this gap reflects the objective and its training recipe rather than data. Spoof-SUPERB shows the same ordering for this pair under an all-layer readout (ITW 21.0 vs. 40.5) [3]. Both English-only encoders remain weak on ASVspoof 5 (38.40 and 41.88 EER; near-degenerate min-DCF). This finding is consistent with Section 4.1: even on this English set, the multilingual encoders are far stronger, and monolingual pretraining transfers poorly regardless of objective. Adding WavLM (masked prediction with a denoising objective, but trained on more and broader data) does not cleanly extend the trend. It improves in-domain and on LA21, while suffering on ITW and ASV5 (Table 1). Thus, a more elaborate objective paired with broader data is not uniformly beneficial and, unlike the wav2vec2/HuBERT pair, confounds objective with data.

4.3. What is and is not backend-robust

Table 2 repeats the study under AASIST, a graph-attention backend architecturally unlike MHFA’s attentive pooling, and we run the same paired bootstrap on it. Several findings hold under both backends. The two higher-coverage encoders (XLS-R, MMS) far outperform the monolingual and 53-language encoders on every OOD set. XLS-R retains the best min-DCF (0.49 vs. MMS 0.60). And

Table 2. AASIST backend: pooled EER (% , mean over 3 seeds) and ASVspoo 5 min-DCF. XLS-R’s cost-metric lead and the coverage-tier gap reproduce under a second, architecturally distinct backend. XEUS shows a seed-dependent collapse under AASIST (high variance on LA21/DF21). [‡]WavLM adds a denoising objective and broader data, so it is not a clean same-data contrast.

Encoder	LA19	LA21	DF21	ITW	ASV5	mDCF
<i>Coverage (matched ~315M, contrastive)</i>						
wav2vec2-LV60 (1)	2.61	23.57	18.73	48.40	38.11	0.93
XLSR-53 (53)	6.00	29.27	19.83	39.61	33.58	0.93
XLS-R (128)	1.75	11.28	9.25	19.17	17.08	0.49
MMS-300M (1406)	2.30	9.32	7.76	23.75	20.97	0.60
<i>Objective (matched LL-60k data, ~315M)</i>						
HuBERT (masked)	3.02	20.56	10.87	33.11	35.96	0.92
WavLM (masked+denoise) [‡]	1.51	21.82	17.95	52.30	40.94	0.89
<i>Reference (577M, outside matched set)</i>						
XEUS (577M ref)	2.95	21.87	24.11	40.72	35.05	0.87

XLS-R’s advantage over MMS on the two *far* sets is significant under both backends (ITW -4.57 [$-5.00, -4.12$]; ASV5 -3.88 [$-3.99, -3.79$]).

On the *near* sets, however, the finer XLS-R-vs-MMS ordering does not merely depend on the backend but significantly reverses. Under AASIST, MMS achieves lower EER on both LA21 and DF21 (by 1.96 and 1.49 points, CIs excluding zero), whereas MHFA favored XLS-R on LA21 and tied on DF21. Thus, the backend determines the winning high-coverage encoder on the near sets, and XLS-R’s lead is robust only on the far OOD sets and the cost metric. Against XEUS, XLS-R’s advantage is significant on all four OOD sets under both backends (more markedly AASIST). AASIST severely destabilizes XEUS as one of three seeds collapses on every benchmark (e.g. LA21 40.6%). Excluding the destabilized seed, the two encoders are closely matched on LA21, so the pooled gap is driven almost entirely by the collapse. We therefore treat MHFA as the primary basis for the XEUS comparison.

4.4. Summary

The two controlled contrasts separate factors that prior comparisons entangled. Out-of-domain generalization improves sharply with coverage up to the ~ 100 -language scale and then plateaus (pushing to 1406 languages, MMS, does not consistently improve EER and worsens the cost metric), showing more coverage is not reliably better. Performance improves under a masked-prediction rather than a contrastive objective on matched data. Neither effect reduces to parameter count, as a 315M encoder matches or surpasses the 577M reference out of domain.

5. DISCUSSION

Why does coverage peak? Our design does not test the mechanism, but the pattern is consistent with a trade-off between breadth and per-language representation quality. XLS-R’s 128-language pretraining plausibly supplies enough phonetic and acoustic diversity to transfer to unseen attacks, while MMS spreads comparable capacity across 1406 mostly low-resource languages, most irrelevant to the English evaluation sets, diluting the representations. Consistent with this, MMS adds to XLS-R’s pretraining corpora ~ 55 K hours of largely single-speaker scripture readings, and its authors report

slight degradation relative to XLS-R on high-resource languages, including English, in fine-tuned ASR [13]. Isolating this mechanism would require controlled pretraining, which public checkpoints do not permit.

Limitations. Three bound our claims. First, across public checkpoints multilingual coverage co-varies with pretraining data *volume* (≈ 50 K to ≈ 500 K hours). We attribute effects to coverage and its correlated data scale, not language count in isolation. Second, all systems are trained on a single corpus (ASVspoo 2019 LA), so our conclusions are scoped to this training regime and the fixed detection recipe. Third, XEUS is a single larger reference on a different encoder lineage, not a matched cell. The observation that a 315M encoder surpasses XEUS out of domain is one data point against scale-as-driver, not a general scaling law.

Pretraining/evaluation provenance. Several evaluation benchmarks share provenance with our encoders’ pretraining corpora: LA19/LA21/DF21 derive substantially from VCTK, which XEUS’s pretraining includes, and ASVspoo 5 is sourced from the English portion of Multilingual LibriSpeech (MLS), a component of XLS-R’s, XLSR-53’s, and MMS’s pretraining mixtures. We cannot fully separate coverage-driven generalization from this exposure on these benchmarks. Two facts bound, but do not fully resolve, the concern. First, XLSR-53 is comparably MLS-exposed yet sits in the low-performing tier of our coverage sweep, so MLS exposure alone does not predict the ranking. Second, MMS-300M’s pretraining includes the same corpora as XLS-R, MLS among them [13], yet XLS-R still leads it on ASVspoo 5, so shared MLS exposure does not account for that margin. However, we cannot rule out that MLS exposure inflates absolute ASVspoo 5 performance for the exposed encoders.

6. CONCLUSION

We presented a controlled decomposition of what drives out-of-domain generalization in frozen SSL encoders for audio deepfake detection. By isolating multilingual coverage at matched capacity and pretraining objective at matched data, we separated two factors that prior comparisons varied together. Within this fixed recipe, coverage helps only up to the ~ 100 -language scale rather than monotonically: a 315M mid-coverage encoder (XLS-R) is the strongest out-of-domain choice on the farther sets, while pushing coverage to 1406 languages does not consistently help. Separately, a masked-prediction objective generalizes better than a contrastive one on identical data. For frozen anti-spoofing, encoder choice should target sufficient but not maximal multilingual coverage and a masked-prediction objective rather than maximal scale. XLS-R emerges as a strong matched-capacity default. Whether these effects generalize across training corpora and detection recipes, and what mechanism produces the coverage peak, are open questions for future work.

Acknowledgments

This work was supported in part by computational resources provided by the NSF ACCESS program under allocation CIS261055. We thank James O’Brien at Sound Ethics for his mentorship. The authors used Claude (Anthropic) for feedback on drafts and suggested revisions to wording in the Abstract, Introduction, Related Work, and Discussion for clarity. All AI-assisted content was reviewed and verified by the authors, who take full responsibility for this manuscript.

7. COMPLIANCE WITH ETHICAL STANDARDS

This work studies detection of, not generation of, synthetic speech, using only publicly available benchmark datasets (ASVspoof 2019/21/5, In-the-Wild) under their respective licenses. No new speech was collected and no human subjects were involved. We release no attack systems. All encoders are publicly available checkpoints and all backends and training configurations are specified in Section 3.

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